

COMP 370 Final Project

Analyzing Joe Biden's Coverage on Reddit During the 2024 Elections

Written by

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Introduction

Joe Biden, the current president of the United States, faced a significant political turning point in 2024 when he chose to withdraw from the presidential race, leaving the Democratic Party to battle with the aftermath of their loss in the election. The 2024 U.S. presidential election marked a pivotal moment in American political discourse, with the party reflecting on the dynamics of media consumption and its implications for voter engagement.

Traditional Democratic-aligned media outlets like MSNBC and CNN saw declines in viewership compared to 2020, while social media platforms, particularly Reddit, became increasingly central to how young voters access news and form opinions (?). Given this shift, our research investigates the portrayal of Joe Biden on Reddit during the election month as it aims to understand how political narratives transform across its different communities.

Using data collected from over 500 Reddit top posts mentioning Joe Biden, we analyzed sentiment (positive, negative, or neutral) and topics within posts across subreddits categorized as Democrat-aligned, Republican-aligned, or mainstream (non-political and general subreddits). The rationale for subreddit selection, sentiment analysis, and topic selection is discussed in the Data and Methods sections. Ultimately, this report aims to answer a specific question: *During election month, do the political opinions mentioning Biden on large mainstream subreddits align more closely with Republican or Democrat-aligned sentiments and topics?*

The findings will shed light on whether mainstream Reddit subreddits represent an opportunity for broad political outreach or lean towards partisan influences. Overall, it assesses the potential reach and influence of political messaging on a platform heavily used by younger voters, who increasingly rely on social media for news.

Why Reddit?

Reddit offers a unique environment for this exploration due to its structure. Users self-select into communities (subreddits) based on interests or affiliations, but popular posts from various subreddits surface on broader feeds, providing a blend of niche and mainstream perspectives. Moreover, the material aspects of anonymity, upvoting, karma, and the rules of each subreddit offer ways in which users can create a 'safe' space to air their views, use convincing language and narrative to recruit others, and reward those who share the same attitudes while excluding alternative views (?). This aspect suggests that mainstream Reddit communities could serve as a viable channel for future advertising and voter outreach for Democratic Campaign managers.

Data

Initially, we were curious about the sentiment towards Biden in right-wing subreddits, and how it differs from the left-leaning ones. However, while collecting the data we realized that it was heavily skewed due to the 'echo-chamber' effect of the subreddits, and it was providing a very distorted viewpoint. Hence, using the Reddit API, we collected data from over 500 posts and labeled them each with a **subreddit political lean**, a **sentiment towards Biden**, and a **topic** discussed. We decided to pull 500 top posts from October 20th, 2024 to November 19th, 2024, intentionally pulling posts surrounding the federal election on November 5th; and avoiding Joe Biden's birthday on November 20th. We collected 167 ($\frac{500}{3}$) posts from each 'cluster': Republican-aligned, Democrat-aligned, and mainstream large subreddits. The following subreddits composed our final dataset:

Democrat-aligned subreddits: [r/antitrumpalliance, r/darkbrandon, r/democrats, r/esist, r/fuckthealtright, r/joe-biden, r/liberal, r/parlerwatch, r/pete.buttigieg, r/republicanvalues, r/sanepolitics, r/the_mueller, r/thebidenshitshow, r/thelib, r/trumpgret, r/trumpvirus, r/votedem, r/wayofthe-bern]

Republican-aligned subreddits: [r/askthe_donald, r/libsofreddit, r/republican, r/thebidenshitshow, r/thetrumpzone, r/timpool, r/trump, r/tucker_carlson, r/conservatives, r/conservativesonly]

Mainstream subreddits: [r/all, r/news, r/pics, r/politics,

To complete the annotation process, the remaining 300 posts were divided among the three group members. However, during a second round of coding, we recognized a bias in topic distribution, likely influenced by the timing of data collection. For instance, the “re-election” category was overly broad and failed to capture sentiment nuances. We addressed this by breaking it into smaller, more specific topics like “re-election campaign”, “voting”, etc. Finally, a third round of open coding ensured all posts were categorized accurately under our finalized 10 topics and 3 sentiment categories.

Data Analysis

Once the articles were coded, a TF-IDF analysis was conducted to identify the prevalent language, and themes present within each category of posts and across sentiments (positive, neutral, negative). The approach was similar to the one taken in Homework 11: Articles were tokenized into words, converted to lowercase, and stripped of non-alphanumeric characters. Then, common stop words (e.g., "the," "and," "for"), as well as the names "Joe" and "Biden," were removed to focus on substantive terms. The inverse document frequency (IDF) was computed across all 500 articles to measure a word's uniqueness relative to the dataset, and the term frequency (TF) was calculated as the number of times a word appeared within a category or sentiment. The 10 words with the highest TF-IDF scores for each category/sentiment were retained for analysis. Lastly, Matplotlib was later used to visualize these results as bar charts.

In addition, we calculated the distribution of all topics and sentiments (positive, negative, neutral) for the entire dataset. The data was then grouped by political leaning, and the same distributions were calculated. Finally, Google Sheets was used to visualize the results as pie charts.

Results

After conducting multiple rounds of coding on the dataset, we developed our typology guide below by refining topics and definitions to ensure consistency in categorization.

Typology Overview

1. Biden-Harris Policies

Posts focusing on the Biden-Harris administration's policies and their impact.

Rules for Inclusion: Posts should address specific policies introduced or supported by the administration, such as Roe v. Wade, border control, climate initiatives, or judicial decisions. Excludes economic policies.

Example from Data: *"Biden administration pushes for stronger climate reforms in 2024" highlights efforts on environmental policy.*

TF-IDF Analysis: Key terms like "roe," "climate," "border," and "judge" highlight the administration's involvement in controversial issues spanning birth control, immigration, and environmental and judicial policies.

2. Economic Impact

Posts addressing economic conditions under Biden's administration.

Rules for Inclusion: Posts must discuss issues like inflation, economic growth, or partisan perspectives on the economy.

Example from Data: *"Biden's relief package helps small businesses amid rising inflation."*

TF-IDF Analysis: Prominent terms such as "inflation" and "economy" reflect widespread inflation concerns, while "republicans" highlight partisan framing and

criticism.

3. Biden's Final Initiatives

Posts focusing on Biden's actions and policies during the final months of his presidency.

Rules for Inclusion: Posts must discuss executive orders, administrative decisions, or legacy-building efforts for November and December.

Example from Data: *"Biden signs executive order to protect voting rights as final act."*

TF-IDF Analysis: Words like "executive," "actions," and "impact" emphasize the focus on Biden's last-minute policy decisions and their potential effects.

4. Foreign Conflict

Posts discussing Biden's foreign policy, particularly related to global conflicts.

Rules for Inclusion: Posts must mention military or diplomatic efforts, such as involvement in Ukraine or relations with Israel.

Example from Data: *"Biden pledges additional aid to Ukraine in response to Russian aggression."*

TF-IDF Analysis: Dominant terms like "ukraine," "russia," "strike," and "missiles" highlight a focus on military action and the conflict in Ukraine. Notably, terms like "palestine" and "israel" are absent, indicating less emphasis on other geopolitical issues.

5. Post-Election

Posts reflecting on the transition of power and Biden's legacy.

Rules for Inclusion: Posts must discuss events related to the post-election period, such as the transfer of power or reflections on Biden's presidency.

Example from Data: *"Biden's legacy secured as he transitions power to Trump with dignity."*

TF-IDF Analysis: Key terms like "white," "house," "legacy," and "jill" emphasize discussions about the transition and First Lady Jill Biden's involvement.

6. Public Well-Being

Posts highlighting governance related to public-facing events and community initiatives.

Rules for Inclusion: Posts must mention social initiatives and activities such as public celebrations, school events, or First Lady initiatives.

Example from Data: *"Jill Biden hosts Halloween event for children at the White House."*

TF-IDF Analysis: Terms like "lady," "halloween," "school," and "event" emphasize a community-oriented narrative centered on public engagement.

7. Re-Election – "Garbage" Comment

Posts discussing the controversy surrounding Biden's "garbage" comment during his campaign.

Rules for Inclusion: Posts must specifically reference the "garbage" comment or its aftermath.

Example from Data: *"Biden clarifies 'garbage' com-*

ment in a speech addressing MAGA supporters.”

TF-IDF Analysis: Words like “garbage,” “supporters,” and “maga” show media focus on the controversy and the reaction from Trump’s base.

8. Re-Election Campaign

Posts related to Biden’s re-election efforts and competition with Trump.

Rules for Inclusion: Posts must mention campaign events, debates, or critiques of opponents.

Example from Data: “Biden rallies Democrats to ‘defend democracy’ against Trump’s agenda.”

TF-IDF Analysis: Terms like “harris,” “trump,” and “election” dominated, indicating a competitive narrative centered on the Biden-Harris campaign.

9. Trump Policies

Posts comparing Biden’s policies to Trump’s.

Rules for Inclusion: Posts must mainly reference Trump’s policy decisions during his term or in contrast to Biden’s governance.

Example from Data: “Trump promise to repeal Biden climate policies could cost US billions, report finds.”

TF-IDF Analysis: Terms like “trump” and “appoint” highlight contrasts in governance styles and decision-making priorities.

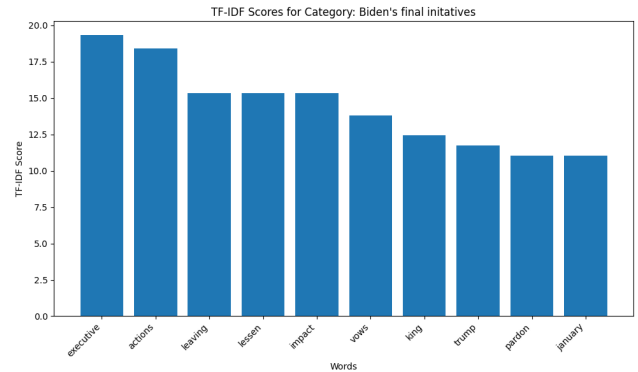
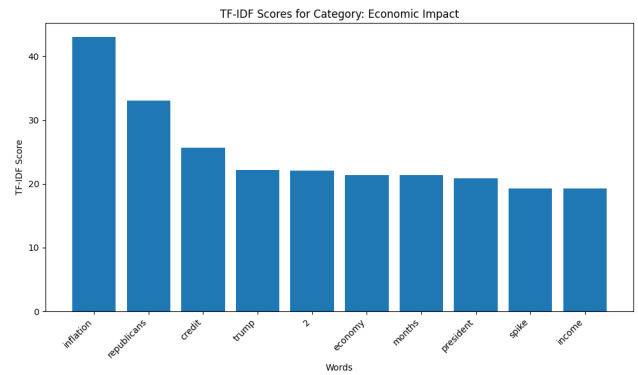
10. Voting

Posts discussing voter behavior, election outcomes, or the act of voting itself.

Rules for Inclusion: Posts must mention voting-related events or data.

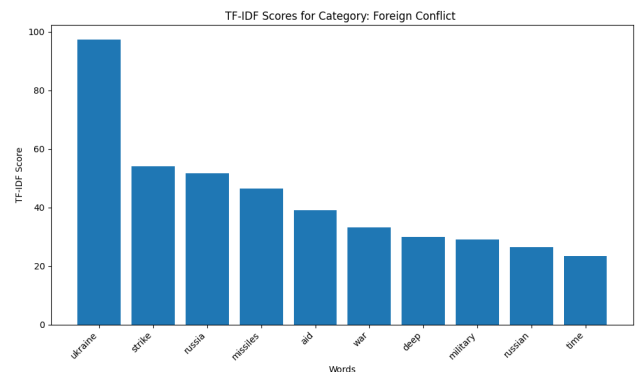
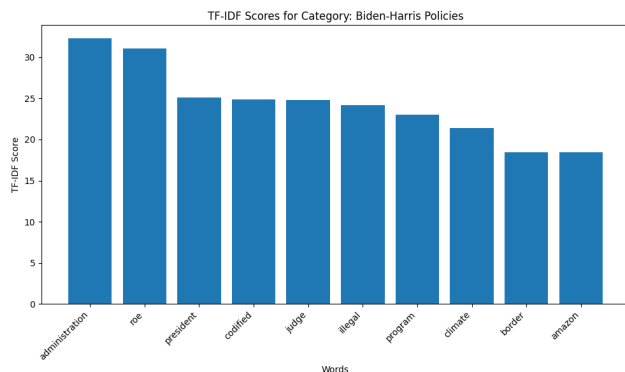
Example from Data: “Who were the 9M Biden 2020 voters who stayed home this year and what are their reasons? 90% did not vote for Trump last week.”

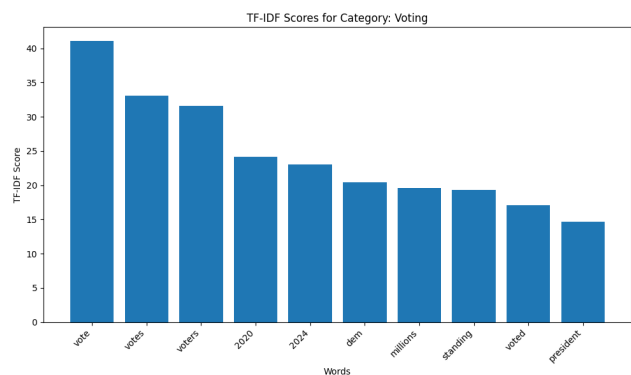
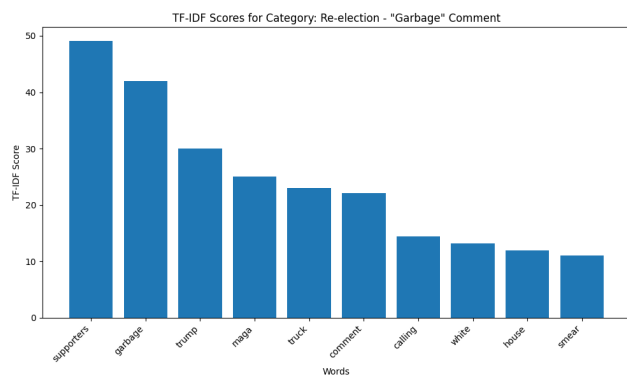
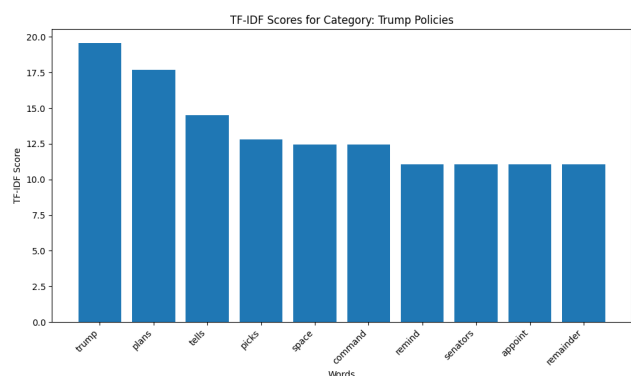
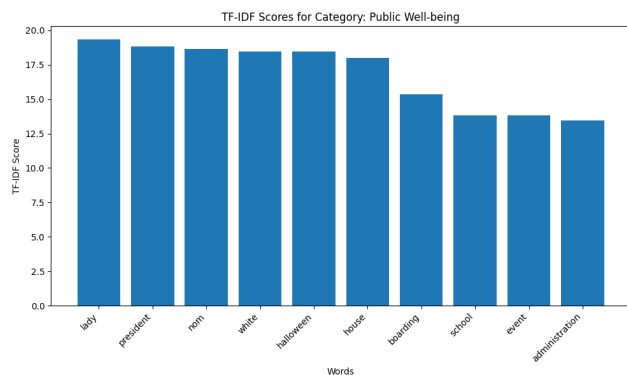
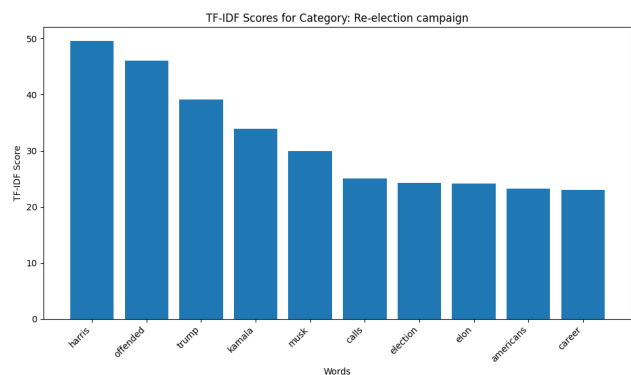
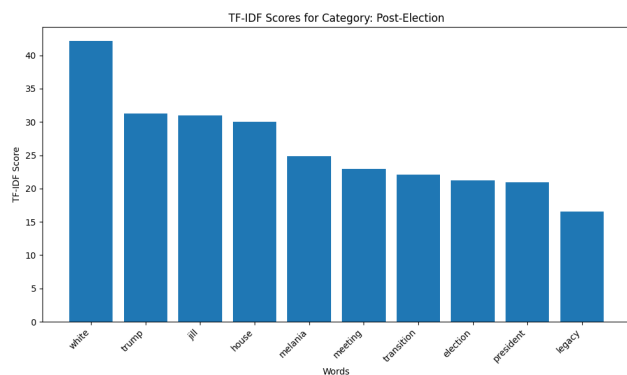
TF-IDF Analysis: Words like “vote,” “voters,” “2020,” and “2024” emphasize discussions on voter engagement and elections past and future.



TF-IDF Results

The top 10 TF-IDF words for each category reveal distinct themes in the media’s focus:





Topic and Sentiment Distribution

Using the typology developed, we annotated the dataset and compiled the topic distributions to compare across different subreddit categories. Figure 2 shows the overall topic distribution across Left-wing, Right-wing, and Mainstream (Neutral) subreddits.

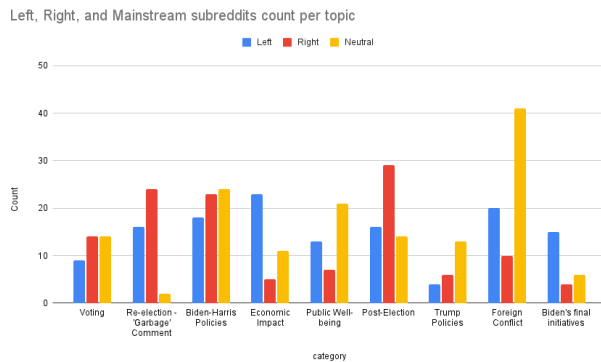


Figure 2: Left, Right, and Mainstream subreddits count per topic

Using the sentiment categories defined earlier, we analyzed the data to explore how sentiments varied across subreddit categories. The overall sentiment distribution is depicted in Figure 6, highlighting the proportions of positive, negative, and neutral sentiments across the entire dataset. Figure 7 presents a bar graph comparing the sentiment distributions within Democrat-aligned, Republican-aligned, and neutral subreddits. Positive sentiments predominantly reflect pro-Joe Biden posts, while negative sentiments are largely associated with pro-Trump narratives, showcasing the polarization within subreddit communities.

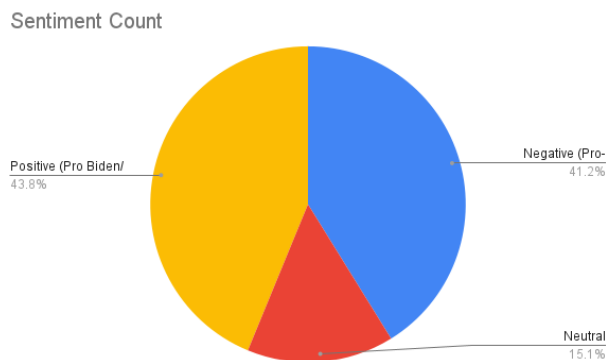


Figure 3: Distribution of Sentiment towards Biden across the dataset.

Discussion

Topics

Figure 2 shows that our highest category of discussion was the re-election campaign, which aligns with the time frame of these posts- before and after the election. During the time before the election, both the democrats and republicans were actively campaigning every single day, giving the media and redditors plenty of quotes to discuss. The second largest category is foreign conflict (14%), which also aligns with assumptions as this was one of the most important

Left, Right, and Mainstream Subreddits and Sentiments

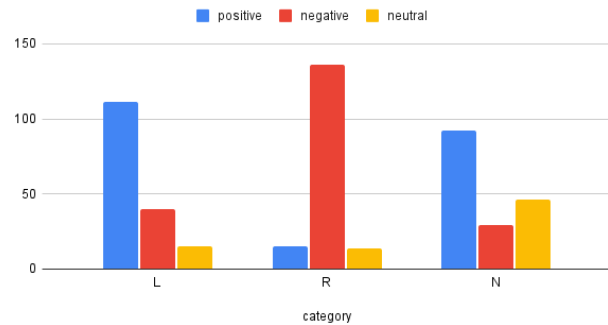


Figure 4: Distribution of Sentiments across Left-wing, Right-wing and Mainstream(Neutral) subreddits.

issues of the current election (Pew Research Center, 2024). At 9% of all topics discussed, the 'Post-Election' reaction is also a large category, and is a majority of republican voters. Since the query used the keyword 'Biden', it makes sense that our lowest category is 'Trump Policies' at about 4%, because posts in this category were typically only a comparison to Biden's administration or policy. According to the Pew Research Center, the economy was the top issue of the election, so it is not surprising that in the analysis we noticed many posts on this topic that we isolated it specifically. At 7.8% of all posts being about these policies in specific, it is clear that our data agrees with the Pew Research Center that this is an important issue, and in regards to Biden, was top of mind for voters this election season. While our data is not every post made about the election, we still saw that in general, the topics discussed in large matches with the key issues about the election. The following will discuss some of the topical outliers in the data

In Figure 2, the majority of posts categorized under "foreign conflict" originate from the mainstream category (42 posts). Upon further examination, most of these posts were sourced from the subreddit r/worldnews, which describes itself as "A place for major news from around the world, excluding US-internal news." This explains the high volume of posts related to foreign affairs within this category. In hindsight, excluding r/worldnews from the analysis of "apolitical" subreddits might have been warranted, given its stated focus on non-US news, which aligns with the rules excluding internal US discussions. However, our results demonstrate that a subreddit's description does not always align with its content or community behavior. For instance, while r/sanepolitics does not explicitly identify as left-leaning in its description, the Reddit map revealed a significant overlap with left-wing subreddits. Our analysis further confirmed this, with 100% of posts from r/sanepolitics being classified as "Pro-Biden." This highlights the importance of examining actual content rather than relying solely on stated descriptions.

In Figure 2, we observe that the "garbage comment"

received minimal discussion within mainstream subreddits. The context of this quote is rooted in Biden's response to a comedian at a Trump rally who joked that Puerto Ricans were "garbage." In reply, Biden stated, "the only garbage I see is his [Trump's] supporters." This controversial remark garnered more attention in smaller, right-leaning communities, as evidenced by posts expressing sentiments like "Biden called us garbage." Given the targeted nature of this statement and its need for contextual understanding, it is unsurprising that mainstream subreddits had limited discussions on the topic, while smaller, ideologically focused communities engaged with it more extensively.

TF-IDF

The analysis of TF-IDF scores reveals distinct categories with high raw values, reflecting key topics in the dataset. Within the Foreign Conflict category, the word "ukraine" stands out with the highest TF-IDF score of 97.42, underscoring the significant media focus on the conflict. This is likely attributed to extensive coverage of recent developments. Other prominent terms, such as "strike" (54.13), "russia" (51.66), and "missiles" (46.50), further highlight the emphasis on military actions and geopolitical tensions. In the Re-Election Campaign category, the word "harris" scores 49.58, indicating a substantial focus on Kamala Harris and her role in the narrative surrounding the campaign. Finally, the Economic Impact category features "inflation" with a score of 43.05, reflecting its importance as a pressing concern for Americans. These high scores provide insight into the primary areas of media and public interest.

Sentiment

Looking at figure 6, positive sentiment, often associated with pro-Biden discussions, dominates in left-leaning subreddits, reflecting a supportive stance among his Democratic base. This sentiment is primarily centered around policy accomplishments and governance. On the other hand, negative sentiment, predominantly pro-Trump, is more prevalent in right-leaning subreddits, indicating strong criticism of Biden and favorable comparisons to Trump. The sentiment analysis reveals an alignment between the online discourse surrounding Joe Biden and the broader political landscape in the United States. The data indicates that 43.8% of sentiments expressed about Biden are positive, closely mirroring Democratic viewpoints, while 41.2% are negative, corresponding largely to Republican perspectives. This distribution reflects the almost even split in political affiliation within the U.S., with Democrats and Republicans holding roughly equal sway over public opinion. This claim is further validated by a survey conducted by Gallup in November 2024, revealing that 47% of the population is Republican or Republican-leaning, while 41% are Democrat or Democrat-leaning, and the rest of the population chooses to identify as independent. These findings suggest that subreddit discourse not only amplifies but also mirrors the polarized nature of American politics, reinforcing the ideological divides prevalent in the nation.

An important factor contributing to this polarization is

the echo chamber effect, particularly evident in Republican-leaning subreddits. As defined by this research: a bias in content moderation creates echo chambers, which are online spaces characterized by homogeneity of opinion and insulation from opposing viewpoints (Godfrey, 2024). Subreddits, by design, create niche spaces where like-minded individuals gather to discuss shared ideologies, often leading to the reinforcement of pre-existing beliefs rather than open debate. In such environments, Republican-leaning subreddits overwhelmingly generate negative sentiments toward Biden, and Democrat-leaning subreddits the opposite, further polarizing discourse. This phenomenon explains why negative or positive sentiments dominate subreddits and naturally lend themselves to self-reinforcing narratives due to the homogeneity of their user base and the minimal intrusion of opposing viewpoints.

Conclusion

This analysis demonstrates how subreddit discourse mirrors the polarized nature of American politics, with sentiments toward Biden aligning closely with broader Democratic and Republican affiliations. The categorization of topics, such as the re-election campaign, foreign conflict, and economic issues, highlights the alignment between public discourse on Reddit and the key issues of the election. Moreover, our findings also reflect on the impact of subreddit-specific dynamics, such as the echo chamber effect and content moderation biases, which reinforce ideological divides and shape sentiment distribution.

However it is noteworthy that the results of this report is limited by time and budget constraints. Thus to improve on this study, future work could expand the dataset to include a broader range of subreddits and a more extended time frame, allowing for a deeper examination of topic evolution and sentiment shifts over time. Additionally, a comparative analysis of other platforms, such as Twitter or Facebook, could provide valuable insights into how Reddit's unique affordances, like anonymity, influence political discourse differently. Finally, incorporating user-level engagement metrics, such as comment frequency or upvote ratios, could further illuminate how discussions propagate within these online communities.

Group Member Contributions

Chloe Seljak: Data collection, open coding, topic creation, data annotation, In the report: Data, Methods (Data Collection), Discussion (Topics).

Rupneet Shahriar: Open coding, topic creation, data annotation, In the report: Creating the report, Introduction, Methods (Annotation), Results, Discussion (Sentiment), and Conclusion.

Thomas Savage-Duguay: Open coding, topic creation, data annotation, TF-IDF Calculation & Charts, In the report: Methods (TF-IDF), Discussion (TF-IDF), Results (TF-IDF)

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